

Madelyn Gatchel

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RESEARCH INTERESTS

I develop methods to study multi-agent strategic interactions using empirical game-theoretic analysis, focusing on learning game models from data, designing algorithms to analyze large games, and applying these techniques to mechanism design.

EDUCATION

University of Michigan

Ph.D. in Computer Science & Engineering

Ann Arbor, MI

September 2021 - Present

- Advisor: Professor Michael P. Wellman, Strategic Reasoning Group

M.S. in Computer Science & Engineering (GPA: 4.00)

August 2023

Davidson College

B.S. in Computer Science (High Honors) and Mathematics, Magna Cum Laude

Davidson, NC

May 2021

- GPA: 3.990 (CS: 4.000; Math: 3.970); Phi Beta Kappa, Omicron Delta Kappa; Thesis advisor: Dr. Bryce Wiedenbeck

AWARDS & HONORS

Rackham Outstanding Graduate Student Instructor Award

Spring 2025

- Recognized for outstanding instruction in EECS 592 AI Foundations (1 of 20 selected university-wide from ~2,000 GSIs).

2nd Place, CSE Graduate Student Honors Competition

Fall 2025

- Recognized for a research presentation in a competition featuring top doctoral students from five CSE research areas.

CSE HACKS Spirit Award

Spring 2024

- Awarded for exemplifying HACKS values: honesty, achievement, cooperation, knowledge, service (1 of ~600 students).

U-M Student Delegate, U7+ Alliance at NEXT Milan Forum

Winter 2024

- One of two U-M delegates; contributed to policy proposals on AI, sustainability, and global access to higher education.

PUBLICATIONS

1. **Madelyn Gatchel** and Michael P. Wellman. Learning Bayesian Game Families, with Application to Mechanism Design. In *25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2026. 2447-2455. **24% acceptance rate.**
2. **Madelyn Gatchel.** Game-Family Learning for Simulation-Based Games (Extended Abstract). In *24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS): Doctoral Consortium*, May 2025. 2929-2931.
3. **Madelyn Gatchel** and Bryce Wiedenbeck. Learning Parameterized Families of Games. In *22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2023. 1044-1052. **23% acceptance rate.**

PRIMARY INSTRUCTOR TEACHING EXPERIENCE

Division of Computer Science & Engineering, University of Michigan

Ann Arbor, MI

EECS 110 Discover Computer Science

Fall 2026

- Expanded course and lecture learning objectives and aligned instruction and assessments.
- Developed PrairieLearn quizzes, a new weekly homework sequence, a course website, and a lesson on generative AI.

EECS 110 Discover Computer Science (31 students; 4.8/5.0 median eval)

Fall 2025

- Developed materials for a new lecture and lab, standardized instructions, and reweighted assignments and problems.
- Organized a student event on academic pathways in computer science beyond EECS 110.

EECS 110 Discover Computer Science (47 students; 4.8/5.0 median eval)

Fall 2024

- Built an assignment tracker, reorganized Canvas course, standardized rubrics, and prepared staff meeting materials.
- F'24, F'25: Interviewed and supervised an undergraduate instructional assistant and a grader.

ADDITIONAL TEACHING EXPERIENCE

Center for Research on Learning and Teaching in Engineering, University of Michigan

Ann Arbor, MI

Engineering Teaching Consultant

August 2025 - Present

- Co-facilitate the practice-teaching portion of undergraduate and graduate TA training (Fall 2025, Winter & Fall 2026).
- Conduct individual teaching consultations for undergraduate and graduate instructors, including mid-semester classroom observations with anonymous student feedback, teaching philosophy reviews, and pedagogical coaching.
- Provost's Seminar on Teaching: Co-facilitated a discussion with faculty on supporting graduate student instructors.

Division of Computer Science & Engineering, University of Michigan

Ann Arbor, MI

Graduate Student Instructor

January 2024 - May 2024

- EECS 592 Artificial Intelligence Foundations (56 students; 4.7/5.0 median eval)

Department of Mathematics & Computer Science, Davidson College

Davidson, NC

Computer Science Assistant Teacher

August 2019 - May 2021

- CSC 121 Programming and Problem Solving (Python): Fall 2019, Spring 2020, Fall 2020 (60-90 students per semester)
 - Spring 2020: Taught three lectures during professor's absence.
 - Fall 2020: Updated quiz handouts, skeleton files, solutions, and autograders to align with restructured course
- CSC/MAT 220 Discrete Structures (introductory proofs course): Spring 2021 (25 students)

DEPARTMENT LEADERSHIP

CSE Ph.D. Admissions Committee, University of Michigan

Ann Arbor, MI

Graduate Student Representative

December 2021 - June 2024

Applicant Recruitment & Support (Fall):

- Created a statement-of-purpose feedback program pairing Ph.D. students with applicants from different backgrounds.
 - 2023: 16 applicants; 2024: 28 applicants, 8 acceptances, and 3 matriculations
- Represented UM CSE at Tapia Conference (2023) and UM CoE EMERGE (2022).
- UM CSE Explore Graduate Studies: Panelist (2022, 2023) and Moderator (2023)

Admissions Committee Efforts:

- Provided a graduate student perspective at weekly CSE Ph.D. admissions committee meetings.
- Designed a rubric system for Ph.D. student reviewers, and recruited and helped train ~50 reviewers (2022; 2023).
- Reviewed ~25 total CSE Ph.D. applications to the AI lab (2021; 2022).
- Evaluated ~20 CSE Ph.D. applications for potential fellowship nomination (2023).

Recruitment Efforts (Winter):

- Helped organize virtual and in-person Ph.D. recruitment weekends (2022-2024).
- Recruited, scheduled, and managed ~50 Ph.D. student volunteers per year for virtual and in-person visits (2023, 2024).

CSE Division Chair Search Committee, University of Michigan

Ann Arbor, MI

Student Representative

July 2023 - December 2023

- Recommended strong prospective candidates, including one who advanced to the final interview stage.
- Conducted first- and second-round interviews with chair candidates and engaged in committee deliberations.
- Organized meetings for graduate students to meet with each second-round chair candidate.

AI Lab, University of Michigan

Ann Arbor, MI

AI Tea Co-Coordinator

December 2021 - May 2023

- Planned and facilitated monthly AI discussions with co-coordinator for 25+ AI lab graduate students and faculty.

Women in Computer Science (WiCS), Davidson College

Davidson, NC

Vice President

August 2019 - May 2021

- Co-organized weekly meetings and co-coordinated event logistics with outside speakers and groups.
- Mentored first-year and sophomore students exploring computer science.

MENTORING

- Steven Zhang, undergraduate research assistant
- Angelina Lee, instructional assistant for EECS 110 (Fall 2025)
- Ena Mestry, instructional assistant for EECS 110 (Fall 2024)

PROFESSIONAL SERVICE

- Workshop reviewer: GAIW-25, GAIW-26
- Conference sub-reviewer: AAMAS-26, ICAIF-25, NeurIPS-24, NeurIPS-23, EC-23
- Conference student volunteer: AAMAS-23, AAMAS-25, AAMAS-26

TEACHING & PEDAGOGICAL DEVELOPMENT

- U-M Graduate Teacher Certificate (Spring 2026)
- Preparing Future Faculty Seminar (Summer 2025)
- U-M Center for Research on Learning and Teaching Workshops:
 - Rubrics: Transparent, Consistent, and Efficient Assessment in Support of Students' Learning (Spring 2024)
 - The Role of Empathy in the Equity-focused Engineering Classroom (Winter 2024)
 - "Are They Getting It?": Low-Stakes Ways to Gauge Student Learning (Winter 2024)
 - Evidence-Based Teaching Practices to Shape Your Response to AI Writing Tools (Spring 2023)

RESEARCH ENGAGEMENT & PROFESSIONAL DEVELOPMENT

- AAMAS Doctoral Consortium Scholar (2025)
- CRA-WP Grad Cohort for Women Scholar (2022)
- AAMAS Student Scholarship (2023, 2026)
- Rackham Graduate School Conference Travel Grant (2023, 2025, 2026)

INSTITUTIONAL SERVICE

Institutional Engagement:

- Michigan Recreation Advisory Committee — Member (September 2024 - Present)
- UM Provost Advisory Committee Meeting — Student Panelist (September 2023)
- CSE Discussions — Presenter and Ph.D. Recruitment Discussion Moderator (October 2023)
- UM College of Engineering Director of Graduate Initiatives Search — Student Interviewer (April 2022)

Student Outreach & Support:

- CSE Undergraduate Research Symposium — Judge (April 2026)
- EECS 601: Introduction to CSE Graduate Research — Student Panelist (November 2022, November 2024)
- Women in Science and Engineering (WISE) Residential Program — Mentor (Fall 2023)
- Summer Research Opportunity Program (SROP) — Graduate Application Mentor (Summer 2023)
- UM Computing CARES Failing Forward — Student Panelist (February 2023)
- EECS 110 Discover Computer Science — Mentor (Winter 2022, Fall 2022)

UNDERGRADUATE HONORS & AWARDS

Research Recognition:

- AAAI-21 Undergraduate Consortium Scholar (Spring 2021)
- CRA-WP Grace Hopper Celebration Research Scholar (Fall 2019)
- DREU Undergraduate Research Assistant for Prof. R. Iris Bahar at Brown University (Summer 2019)

Academic Honors:

- Davidson College Senior Computer Science Award (Spring 2021)
- Davidson College William Gillespie McGavock Mathematics Award (Spring 2021)
- Davidson College Bernard Society of Mathematics Sophomore Computer Science Award (Fall 2019)

Leadership & Service:

- Davidson College Residence Life Office Award (Spring 2019)
- Davidson College Chidsey Leadership Fellow (2018-2021)